Shanghai SHSID

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A Self-adaptive Multi-terrain Bionic Ankle

Problem

Robots that focuses on movement have always been a top favorite in the creation of robotics. This started from build wheel robots, that can quickly navigate smooth surfaces, namely human-built pathways. Then it went to building leg-powered robots, which requires a good ankle to navigate the rough, wild terrains.

Background

The idea of a bionic ankle have existed for ages, but most bionic ankle focuses on improving only the gait pattern of the bionic ankle and shock absorption.

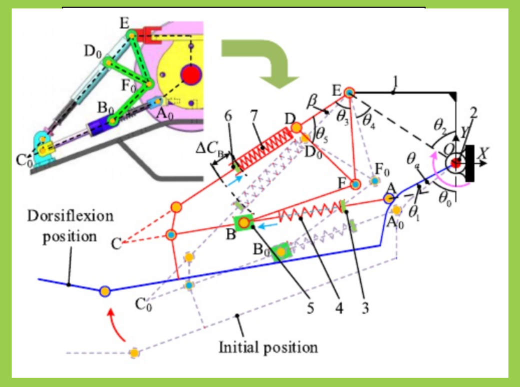


Figure 1: A Multi-loop Ankle Foot

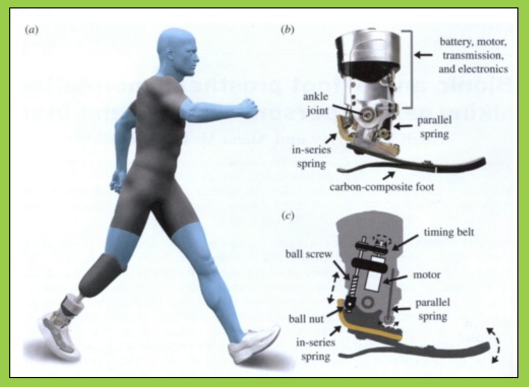


Figure 2: BiOM's bionic ankle

Innovation

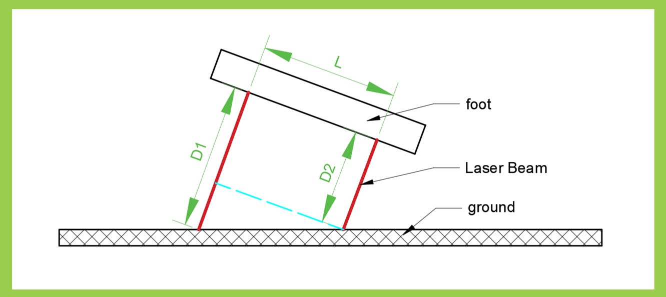
1. Added inversion and eversion movements
2. Used a better spring system to mimic human's ankle function
3. Created a self-adaptive program that uses a combination of IMU and laser distance sensors to aid walking on uneven ground.

Function

The bionic ankle will use two motors to mimic the movement of a normal human foot. Using the components spring system, universal joint, and motor system, the bionic foot tries to mimic the body part's function, including

1. abosorbing shock,
2. providing power,
3. providing joints
4. providing supports

Distance Sensors



Detecting the ground is one of the important objective in the self-adpative program.

Using the Distance sensors, the ankle can use trignometry to find out the angle between using the equation $\angle A = \tan^{-1}\frac{\Delta d}{l}$. Since one distance sensors cannot fully represent the ground, each end's distance is the average of the 2 distance sensors on that end in order to represent the whole plane.

Solution

I am trying to make a bionic ankle that can relatively mimic a human ankle, and help disabled people to walk once more.

1. Using Solidworks, I first plan out the whole structure of hardware to check out the plan's feasibility.
2. Then, I started on building the whole project's hardware.

a) The supporting system is carefully considered to be close to an actual adult human lower leg

Figure 5 (left): Bionic Ankle [side view]
Figure 6 (right): Bionic Ankle [back view]

a) The motors are attached to axles, along with track rollers and flange bearings to decrease friction and attaching to supporting boards

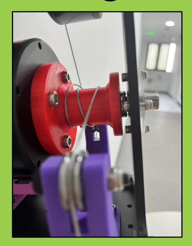
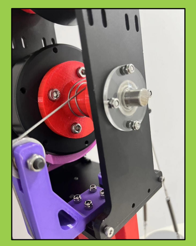


Figure 7 (left): Motor, Axle, Track rollers
Figure 8 (right): Flange Bearing



b) There are then guide wheels to guide the strings to their designated places and damping to reduce the vibration of the motors.

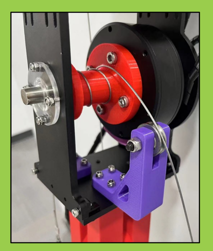
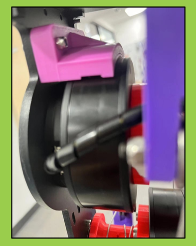
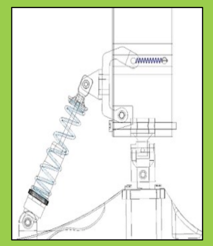


Figure 9 (left): Guide wheels
Figure 10 (right): Damping



c) The spring system contains of two parallel groups each with two springs so that the springs can apply to all directions

Figure 11 (left): Spring system [model]
Figure 12 (right):Spring system [transparent]



d) The universal joint is connected between the foot and the leg. The parts connecting the universal joint are specially designed to withstand power.

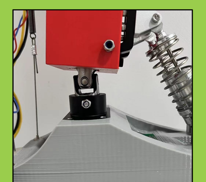
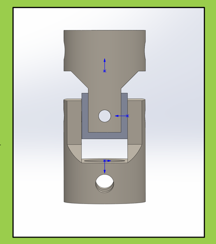
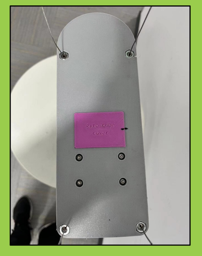


Figure 13 (left): Universal Joint [in model]
Figure 14 (right):Universal Joint



e) The strings are connected to springs to give it more stretch, and fastened tightly to the ends of the foot, wrapping around on the motors

Figure 15 (left): Springs in strings
Figure 16 (right): Foot bottom, with strings binding to it



3. I then build the Arduino setup, using esp32 as the controller, a gyroscope as a sensor, and two motors as the drive force. Esp32 can wirelessly send data to other esp32. This means the gyroscope, which is embedded into the foot, do not need to connect a wire all the way down.

Programming

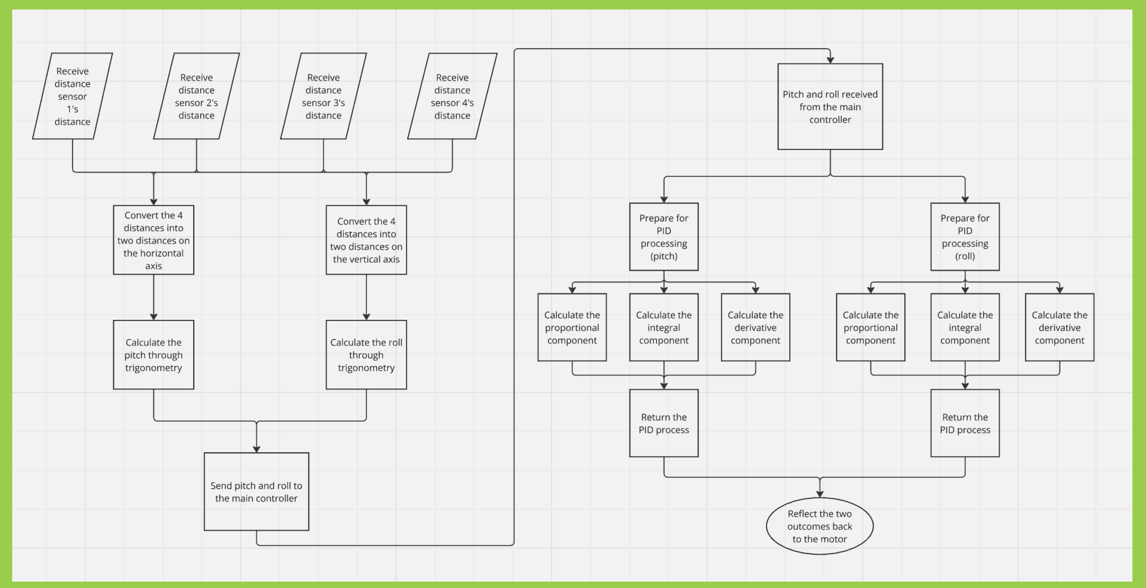


Figure 16: Software Flowchart

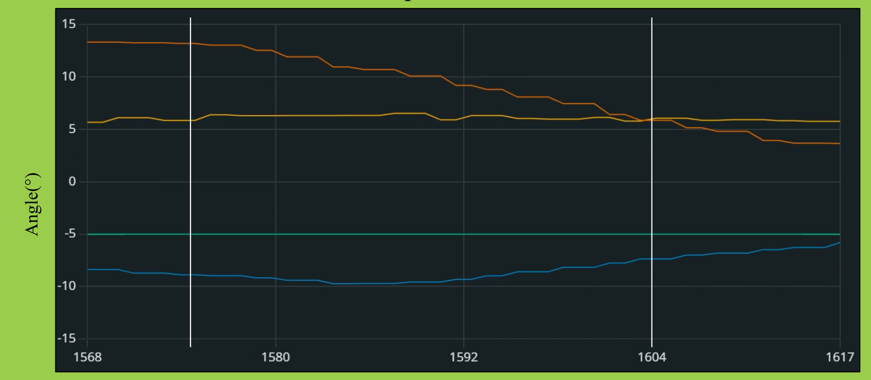
```
float PID_control2 (float now_date2,float set_date2)
{
  float err2 = set_date2 - now_date2;
  err_all2 = err2 + err_all2;
  if (err_all2 > 1000) err_all2 = 1000;
  if (err_all2 < -1000) err_all2 = -1000;
  float P_OUT2 = p_kp * err2;
  float I_OUT2= p_ki * err_all2;
  float D_OUT2 = p_kd * (err2 - P_last_err);
  P_last_err = err2;
  float all_out2 = P_OUT2 + I_OUT2 + D_OUT2;
  if (all_out2 >1200) all_out2 = 1200;
  if (all_out2 < -1200) all_out2 = -1200;
  return all_out2;

}

void PID_total (float roll, float pitch, float set_pitch, float set_roll) {
  roll_PID_angle = PID_control(roll, set_roll);
  pitch_PID_angle = PID_control2(pitch, set_pitch);
  angle_increase_send(1,roll_PID_angle - pitch_PID_angle); delay(10);
  angle_increase_send(2,-roll_PID_angle - pitch_PID_angle); delay(5);
}
```

Figure 17: PID controller code

PID_control and PID_control2 are the same function using different variables

The programming of the bionic ankle relies on a 2 variable PID control. The four distance sensors will calculate the inclination of the foot relative to the ground. Then, the self adaptive program will separately apply PID controller on the two element. Then, combine the two outcomes and send it to the two motors, letting them increase or decrease their angle.

Graph 1: PID control outcome over time



Color	Meaning
Red	Actual pitch
Yellow	Desired pitch
Green	Desired roll
Blue	Actual roll

Time interval(ms)

From 1575 ms to 1600 ms, in this 25 ms time period, the PID algorithm reacted quickly to approach their desired pitch value and desired roll value.

Final Product

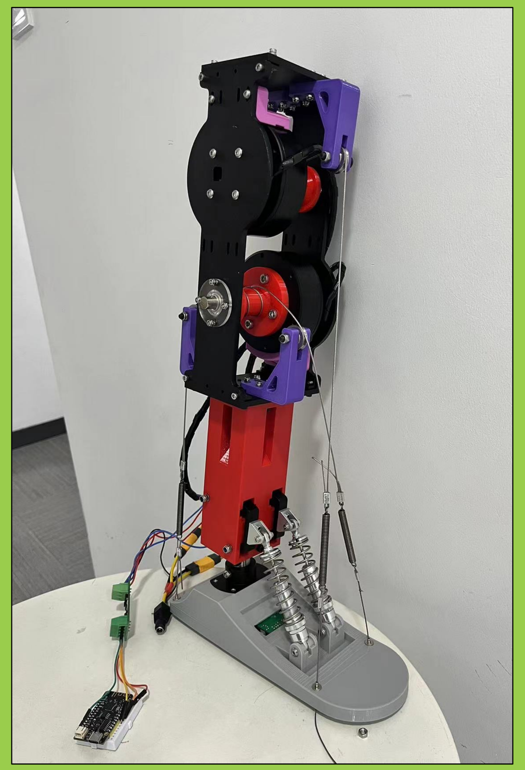


Figure 18: Actual Work

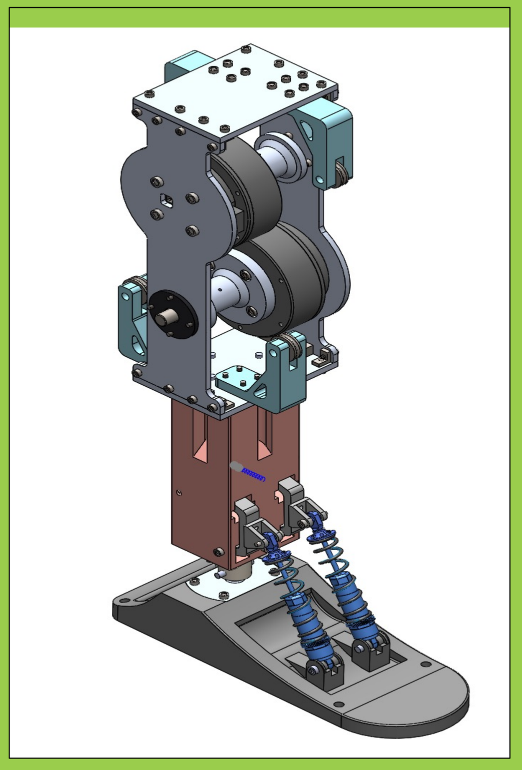


Figure 19: Solidworks Model

Experiment 1-Ankle's movement

This experiment attempts to explore how much can the foot move in the four directions: plantar flexion, dorsiflexion, inversion, and eversion.

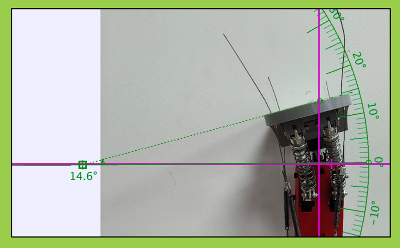


Figure 20: Angle of Eversion

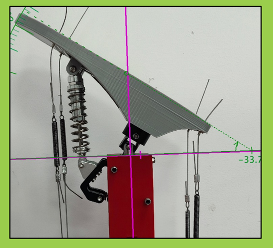


Figure 21: Angle of Planar Flexion

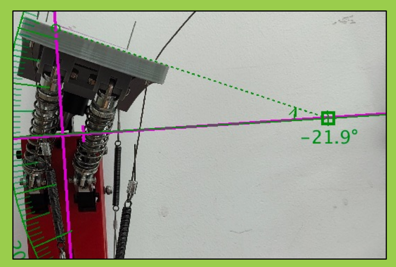
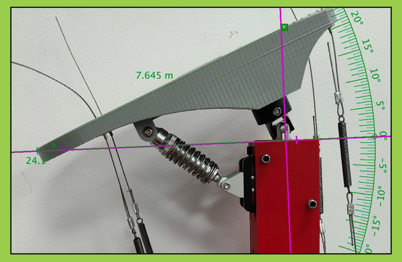


Figure 22: Angle of Inversion



23: Angle of Dorsiflexion

Motion	Maximum angle(°)
Dorsiflexion (up)	24.1
Plantar flexion (down)	33.7
Inversion (inward)	21.9
Eversion (outward)	14.6

Table 1: The maximum angle that each movement can get

These numbers relatively matches the human's ankle movements, with the percent accuracy, except dorsiflexion, are around 26%. This means that the ankle can exceed to human ability, and even surpress it.

Experiment 2-Adaption to ground

This experiment shows the bionic ankle's adaptation to the ground. In the two diagram, the block acts like the ground. The foot will follow the ground and move according to the block.

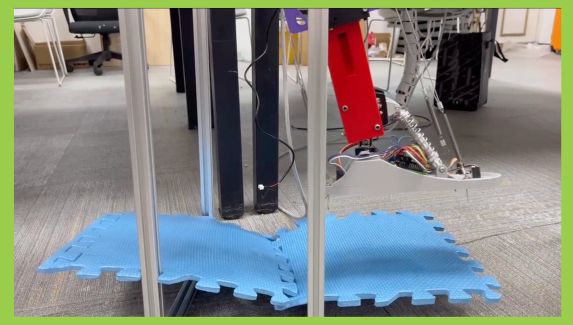
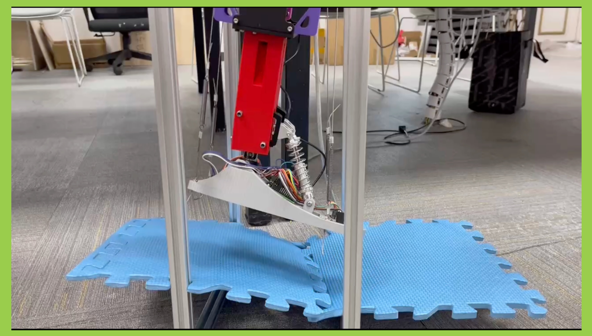


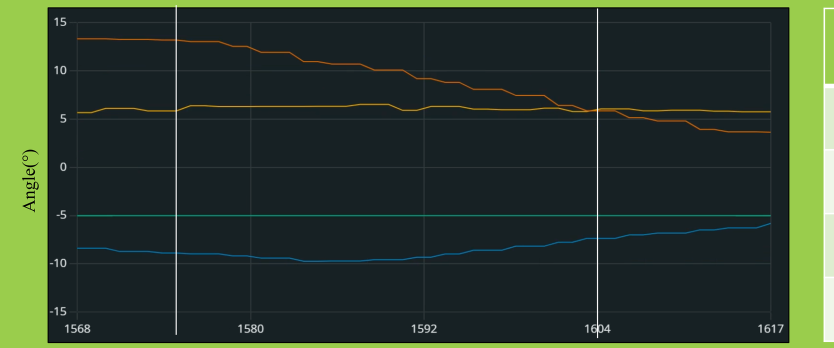
Figure 24: Before Time slice of adaptation to the ground video

Figure 25: After Time slice of adaptation to the ground video



The below graph is the values of the experiment. As the block moves, the desired pitch and roll will change, therefore the foot's pitch and roll will move according to it.

Graph 1 clone: PID control outcome over time



Color	Meaning
Red	Actual pitch
Yellow	Desired pitch
Green	Desired roll
Blue	Actual roll

Time interval(ms)

This adaption to the ground can help the users tranverse complex terrains and environment.